Noam Brown

Curriculum Vitae

Education

2014–2020 **PhD in Computer Science**, *Carnegie Mellon University*
Advisor: Tuomas Sandholm
School of Computer Science Distinguished Dissertation Award

2012–2014 **MS in Robotics**, *Carnegie Mellon University*
Advisor: Tuomas Sandholm

2005–2009 **BA in Mathematics and Computer Science**, *Rutgers University*
Summa Cum Laude
Rutgers College Honors Program

Work Experience

2023–Present **Member of Technical Staff**, *OpenAI*, San Francisco

2018–2023 **Research Scientist**, *Facebook AI Research*, New York

2012–2018 **Research Assistant**, *Carnegie Mellon University*, Pittsburgh

Summer 2017 **Research Intern**, *DeepMind*, London

2010–2012 **Research Assistant**, *Federal Reserve Board of Governors*, Washington, DC

2006–2010 **Algorithmic Trading Engineer**, *MJM Trading Group*, New York

Publications

Journal Papers

- Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning. Meta Fundamental AI Research Diplomacy Team (FAIR), Anton Bakhtin*, Noam Brown*, Emily Dinan*, Gabriele Farina, Colin Flaherty*, Daniel Fried, Andrew Goff, Jonathan Gray*, Hengyuan Hu*, Athul Paul Jacob*, Mojtaba Komeili, Karthik Konath, Minae Kwon, Adam Lerer*, Mike Lewis*, Alexander H. Miller*, Sasha Mitts, Adithya Renduchintala, Stephen Roller, Dirk Rowe, Weiyan Shi*, Joe Spisak, Alexander Wei, David Wu*, Hugh Zhang*, Markus Zijlstra. *Science*, 2022.
- Superhuman AI for Multiplayer Poker. Noam Brown and Tuomas Sandholm. *Science*, 2019. **Cover Article.**
- Superhuman AI for Heads-Up No-Limit Poker: Libratus Beats Top Professionals. Noam Brown and Tuomas Sandholm. *Science*, 2017.

Technical Reports

- OpenAI GPT-5 System Card. OpenAI (including Noam Brown). Technical report, 2025. arXiv:2601.03267.
- OpenAI o1 System Card. OpenAI (including Noam Brown). Technical report, 2024. arXiv:2412.16720.

Conference Papers

- A Regularization-Based Approach to Public Belief State Search for Adversarial Games. Sobhan Mohammadpour, Samuel Sokota, Brandon Gary Kaplowitz, J. Zico Kolter, Noam Brown, Gabriele Farina. Neural Information Processing Systems (NeurIPS), 2026. To appear.
- The Update-Equivalence Framework for Decision-Time Planning. Samuel Sokota, Gabriele Farina, David J. Wu, Hengyuan Hu, Kevin A. Wang, J. Zico Kolter, Noam Brown. International Conference on Learning Representations (ICLR), 2024.
- Abstracting Imperfect Information Away from Two-Player Zero-Sum Games. Samuel Sokota, Ryan D'Orazio, Chun Kai Ling, David J. Wu, J. Zico Kolter, Noam Brown. International Conference on Machine Learning (ICML), 2023.
- Mastering the Game of No-Press Diplomacy via Human-Regularized Reinforcement Learning and Planning. Anton Bakhtin*, David J Wu*, Adam Lerer*, Jonathan Gray*, Athul Paul Jacob*, Gabriele Farina*, Alexander H Miller, Noam Brown. International Conference on Learning Representations (ICLR), 2023. **Outstanding Paper Honorable Mention.**
- A Unified Approach to Reinforcement Learning, Quantal Response Equilibria, and Two-Player Zero-Sum Games. Samuel Sokota*, Ryan D'Orazio*, J. Zico Kolter, Nicolas Loizou, Marc Lanctot, Ioannis Mitliagkas, Noam Brown, Christian Kroer. International Conference on Learning Representations (ICLR), 2023.
- Modeling Strong and Human-Like Gameplay with KL-Regularized Search. Athul Paul Jacob*, David J Wu*, Gabriele Farina*, Adam Lerer, Hengyuan Hu, Anton Bakhtin, Jacob Andreas, Noam Brown. International Conference on Machine Learning (ICML), 2022.
- A Fine-Tuning Approach to Belief State Modeling. Sam Sokota, Hengyuan Hu, David J Wu, Zico Kolter, Jakob Foerster, Noam Brown. International Conference on Learning Representations (ICLR), 2022.
- Equilibrium Finding in Matrix Games Via Greedy Regret Minimization. Hugh Zhang, Adam Lerer, Noam Brown. AAAI Conference on Artificial Intelligence (AAAI), 2022.
- Scalable Online Planning via Reinforcement Learning Fine-Tuning. Arnaud Fickinger*, Hengyuan Hu*, Brandon Amos, Stuart Russell, Noam Brown. Neural Information Processing Systems (NeurIPS), 2021.
- No-Press Diplomacy from Scratch. Anton Bakhtin, David J Wu, Adam Lerer, Noam Brown. Neural Information Processing Systems (NeurIPS), 2021.
- Human-Level Performance in No-Press Diplomacy via Equilibrium Search. Jonathan Gray*, Adam Lerer*, Anton Bakhtin, Noam Brown. International Conference on Learning Representations (ICLR), 2021.
- Off-Belief Learning. Hengyuan Hu, Adam Lerer, Brandon Cui, Luis Pineda, Noam Brown, Jakob Foerster. International Conference on Machine Learning (ICML), 2021.
- Safe Search for Stackelberg Equilibria in Extensive-Form Games. Chun Kai Ling, Noam Brown. AAAI Conference on Artificial Intelligence (AAAI), 2021.
- Combining Deep Reinforcement Learning and Search for Imperfect-Information Games. Noam Brown*, Anton Bakhtin*, Adam Lerer, Qucheng Gong. Neural Information Processing Systems (NeurIPS), 2020.

- Improving Policies via Search in Partially Observable Games. Adam Lerer, Hengyuan Hu, Jakob Foerster, Noam Brown. AAAI Conference on Artificial Intelligence (AAAI), 2020.
- Deep Counterfactual Regret Minimization. Noam Brown*, Adam Lerer*, Sam Gross, Tuomas Sandholm. International Conference on Machine Learning (ICML), 2019.
- Stable-Predictive Optimistic Counterfactual Regret Minimization. Gabriele Farina, Christian Kroer, Noam Brown, Tuomas Sandholm. International Conference on Machine Learning (ICML), 2019.
- Solving Imperfect-Information Games via Discounted Regret Minimization. Noam Brown and Tuomas Sandholm. AAAI Conference on Artificial Intelligence (AAAI), 2019. **Outstanding Paper Honorable Mention.**
- Depth-Limited Solving for Imperfect-Information Games. Noam Brown, Tuomas Sandholm, Brandon Amos. Neural Information Processing Systems (NeurIPS), 2018.
- Safe and Nested Subgame Solving for Imperfect-Information Games. Noam Brown and Tuomas Sandholm. Neural Information Processing Systems (NeurIPS), 2017. **Best Paper Award.**
- Reduced Space and Faster Convergence in Imperfect-Information Games via Pruning. Noam Brown and Tuomas Sandholm. International Conference on Machine Learning (ICML), 2017.
- Dynamic Thresholding and Pruning for Regret Minimization. Noam Brown, Christian Kroer, Tuomas Sandholm. AAAI Conference on Artificial Intelligence (AAAI), 2017.
- Strategy-Based Warm Starting for Regret Minimization in Games. Noam Brown and Tuomas Sandholm. AAAI Conference on Artificial Intelligence (AAAI), 2016.
- Regret-Based Pruning in Extensive-Form Games. Noam Brown and Tuomas Sandholm. Neural Information Processing Systems (NeurIPS), 2015.
- Simultaneous Abstraction and Equilibrium Finding in Games. Noam Brown and Tuomas Sandholm. International Joint Conference on Artificial Intelligence (IJCAI), 2015.
- Hierarchical Abstraction, Distributed Equilibrium Computation, and Post-Processing, with Application to a Champion No-Limit Texas Hold'em Agent. Noam Brown*, Sam Ganzfried*, Tuomas Sandholm. International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2015.
- Regret Transfer and Parameter Optimization. Noam Brown and Tuomas Sandholm. AAAI Conference on Artificial Intelligence (AAAI), 2014.

Workshop Papers

- Human-AI Coordination via Human-Regularized Search and Learning. Hengyuan Hu, David J. Wu, Adam Lerer, Jakob Foerster, Noam Brown. NeurIPS Deep RL Workshop, 2022.
- Learned Belief Search: Efficiently Improving Policies in Partially Observable Settings. Hengyuan Hu*, Adam Lerer*, Noam Brown, Jakob Foerster. AAAI Workshop on Reinforcement Learning in Games, 2021.
- DREAM: Deep Regret Minimization with Advantage Baselines and Model-Free Learning. Eric Steinberger, Adam Lerer, Noam Brown. NeurIPS Deep RL Workshop, 2021.
- Unlocking the Potential of Deep Counterfactual Value Networks. Ryan Zarick, Bryan Pellegrino, Noam Brown, Caleb Banister. NeurIPS Deep RL Workshop, 2021.

Refereed Demonstrations

- Libratus: Beating Top Humans in No-Limit Poker. Noam Brown and Tuomas Sandholm. Demonstrations Program at Neural Information Processing Systems (NeurIPS), 2017.
- Libratus: The Superhuman AI for No-Limit Poker. Noam Brown and Tuomas Sandholm. Demonstrations Program at International Joint Conference on Artificial Intelligence (IJCAI), 2017.

- Baby Tartanian8: Winning Agent from the 2016 Annual Computer Poker Competition. Noam Brown and Tuomas Sandholm. Demonstrations Program at International Joint Conference on Artificial Intelligence (IJCAI), 2016.
- Claudico: The World's Strongest No-Limit Texas Hold'em Poker AI. Noam Brown and Tuomas Sandholm. Demonstrations Program at Neural Information Processing Systems (NeurIPS), 2015.
- Tartanian7: A Champion Two-Player No-Limit Texas Hold'em Poker-Player Program. Noam Brown, Sam Ganzfried, Tuomas Sandholm. Demonstrations Program at AAAI Conference on Artificial Intelligence (AAAI), 2015.

Awards

2023 ICLR Outstanding Paper Honorable Mention
2020 AAAI ACM-SIGAI Dissertation Award
2020 IFAAMAS Victor Lesser Dissertation Award
2020 Carnegie Mellon School of Computer Science Distinguished Dissertation Award
2019 Pluribus chosen as one of nine runners-up for Science Magazine's Breakthrough of the Year
2019 MIT Tech Review 35 Innovators Under 35
2019 Marvin Minsky Medal for Outstanding Achievements in AI
2019 AAAI Outstanding Paper Honorable Mention (one of four papers receiving special recognition out of 7,095 submissions and 1,150 accepted papers)
2018 Open Philanthropy AI Fellowship (one of seven recipients)
2018 Tencent AI Lab Fellowship (one of five recipients)
2018 Libratus won HPCWire's "Best Use of AI" Award (again)
2017 NeurIPS Best Paper Award (one of three out of 3,240 submissions and 678 accepted papers)
2017 Allen Newell Award for Research Excellence
2017 Libratus (together with DeepStack) selected as one of 12 candidates for Science Magazine's Scientific Breakthrough of the Year
2017 Libratus listed in La Recherche as one of the top ten scientific achievements of 2017
2017 Libratus won HPCWire's "Best Use of AI" Award
2016 1st place in the Annual Computer Poker Competition No-Limit Texas Hold'em Instant Runoff Event (there was no 2015 competition)
2016 1st place in the Annual Computer Poker Competition No-Limit Texas Hold'em Total Bankroll Event (there was no 2015 competition)
2014 1st place in the Annual Computer Poker Competition No-Limit Texas Hold'em Instant Runoff Event
2014 1st place in the Annual Computer Poker Competition No-Limit Texas Hold'em Total Bankroll Event
2014 NSF Graduate Research Fellowship Honorable Mention
2009 Rutgers Computer Science Department Highest Honors
2005–2009 Rutgers College Scholarship Recipient

Invited Talks

2026

- **Implications of Large-Scale Test-Time Compute**
  FAR.AI Seoul Alignment Workshop. Lightning talk. July 6, 2026.
- **Implications of Large-Scale Test-Time Compute**
  Industry keynote, Global AI Frontiers Symposium, Seoul. July 3, 2026.
- **Implications of Large-Scale Test-Time Compute**
  Manifest, Berkeley. June 14, 2026.
- **RL for LLMs: Reasoning**
  CS224R Deep Reinforcement Learning, Stanford University. Guest lecture. May 1, 2026.
- **Building a Superhuman AI for No-Limit Poker**
  MIT Pokerbots. Guest lecture (virtual). January 28, 2026.

2025

- **Toward Scientific Discovery: Lessons from an AI's IMO Gold Performance**
  Stanford NLP Seminar. October 23, 2025.
- **Fireside Chat with Joe Spisak**
  Measuring Intelligence Summit at PyTorch Conference, San Francisco. Opening keynote fireside chat. October 21, 2025.
- **Multi-Agent AI**
  CS294/194-196 Agentic AI, University of California, Berkeley. Guest lecture. October 20, 2025.
- **AI for Science Bottlenecks**
  Progress Conference, Berkeley. Panel discussion. October 17, 2025.
- **Learning to Reason with LLMs**
  Manifest, Berkeley. June 2025.
- **Learning to Reason with LLMs**
  National University of Singapore. April 28, 2025.
- **High-Compute Alignment & Control**
  FAR.AI Singapore Alignment Workshop. Presented on behalf of Stephen McAleer. April 23, 2025.
- **Advancing AI Reasoning: From Games to Complex Problem Solving**
  NVIDIA GTC, San Jose. Panel discussion. March 19, 2025.
- **Thinking Machines: How Reasoning AI Is Rewriting the Future of Work, Science, and Strategy**
  AI Economics in the OpenAI Forum, San Francisco. March 5, 2025.
- **Learning to Reason with LLMs**
  Kempner Institute, Harvard University. January 31, 2025.
- **Learning to Reason with LLMs**
  MIT CSAIL. January 30, 2025.
- **Building a Superhuman AI for No-Limit Poker**
  MIT Pokerbots. Guest lecture. January 29, 2025.

2024

- **Learning to Reason with LLMs**
  MATH-AI: Workshop on Mathematical Reasoning and AI, NeurIPS, Vancouver. December 14, 2024.
- **Beyond Decoding: Meta-Generation Algorithms for Large Language Models**
  NeurIPS tutorial, Vancouver. Panel participant. December 10, 2024.
- **Learning to Reason with LLMs**
  SCS Katayanagi Distinguished Lecture, Carnegie Mellon University. November 21, 2024.
- **Learning to Reason with LLMs**
  RAIN/OR Seminar, Stanford University. November 18, 2024.
- **Fireside Chat with Brad Gerstner**
  Altimeter Investor Day, Menlo Park. October 8, 2024.
- **AI Won't Plateau — If We Give It Time to Think**
  TEDAI San Francisco. October 2024.
- **Learning to Reason with LLMs**
  Simons Institute, University of California, Berkeley. September 26, 2024.
- **Parables on the Power of Planning in AI: From Chess to Natural Language Diplomacy and Beyond**
  TNG Big Techday 24, Munich. July 5, 2024.
- **Self-Play**
  Cooperative AI Summer School, Santa Cruz. Remote lecture. June 21, 2024.
- **Parables on the Power of Planning in AI: From Poker to Diplomacy**
  Paul G. Allen School Distinguished Lecture, University of Washington. May 23, 2024.
- **A Parable on the Power of Planning**
  Sequoia AI Ascent, San Francisco. March 20, 2024.
- **Building a Superhuman AI for No-Limit Poker**
  MIT Pokerbots. Guest lecture. January 29, 2024.

2023

- **Guest Lecture on Diplomacy**
  15-888 Computational Game Solving, Carnegie Mellon University. Guest lecture (virtual). November 8, 2023.
- **CICERO: Human-Level Performance in the Game of Diplomacy by Combining Language Models with Strategic Reasoning**
  Information Systems Laboratory Colloquium, Stanford University. October 19, 2023.
- **AI Panel and Roundtable**
  Sequoia / Founders Forum FF100, Switzerland. September 30, 2023.
- **Learning to Cooperate and Compete via Self-Play**
  Future of Decentralization, AI, and Computing Summit, Berkeley RDI, University of California, Berkeley. August 27, 2023.
- **CICERO: Human-Level Performance in the Game of Diplomacy by Combining Language Models with Strategic Reasoning**
  International Joint Conference on Artificial Intelligence (IJCAI), Macao. Keynote. August 24, 2023.
- **Learning to Cooperate and Compete via Self-Play**
  Cooperative AI Summer School, London. July 18, 2023.

- **AI/ML Fireside Chat**
  Sequoia Capital. Fireside chat (virtual). July 7, 2023.
- **CICERO: Learning to Cooperate and Compete with Humans in the Negotiation Game of Diplomacy**
  CHAI Annual Workshop, Asilomar, California. Invited talk and Cooperative AI panel. June 17, 2023.
- **A Parable on the Power of Planning: From Poker to Diplomacy**
  Graduate AI course, University of Massachusetts Amherst. Guest lecture (virtual). March 7, 2023.
- **Strategic Games**
  CS25: Transformers United V2, Stanford University. Guest lecture. January 31, 2023.
- **Building a Superhuman AI for No-Limit Poker**
  MIT Pokerbots. Guest lecture. January 18, 2023.

2022

- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  University of Oxford. November 2022.
- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  XTX. November 2022.
- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  UT Austin. October 2022.
- **Learning to Cooperate and Compete in Diplomacy**
  Flatiron Institute. June 2022.
- **Learning to Cooperate and Compete in Diplomacy**
  Tel Aviv University. March 2022.
- **Learning to Cooperate and Compete in Diplomacy**
  Bar Ilan University. March 2022.
- **Learning to Cooperate and Compete in Diplomacy**
  Hebrew University. March 2022.
- **Learning to Cooperate and Compete in Diplomacy**
  Technion. March 2022.
- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  University of California, Berkeley, Simons Institute Workshop on Adversarial Approaches in Machine Learning. February 2022.
- **Tutorial on AI for Imperfect-Information Games**
  AAAI Reinforcement Learning in Games Workshop. February 2022.

2021

- **Learning to Cooperate and Compete in Diplomacy**
  Stanford University. December 2021.
- **Learning to Cooperate and Compete in Diplomacy**
  University of California, Berkeley. December 2021.

- **Learning to Cooperate and Compete in Diplomacy**
  Harvard University. September 2021.
- **Learning to Cooperate and Compete in Diplomacy**
  MIT. September 2021.
- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  London Machine Learning Meetup. May 2021.
- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  University of California, Berkeley. April 2021.
- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  University of California, San Diego. January 2021.

2020

- **ReBeL: Combining Deep Reinforcement Learning and Search for Imperfect-Information Games**
  University of Michigan. August 2020.
- **Superhuman AI for Multiplayer Poker**
  Conference on Economics and Computation, Highlights Beyond EC. July 2020.
- **Superhuman AI for Multiplayer Poker**
  Johns Hopkins University Applied Physics Laboratory. June 2020.
- **Superhuman AI for Multiplayer Poker**
  University of California, Berkeley. May 2020.

2019

- **Tutorial on AI for Imperfect-Information Games**
  International Conference on Distributed AI (DAI). October 2019.
- **AI for Imperfect-Information Game Settings**
  Tsinghua University. October 2019.
- **AI for Imperfect-Information Game Settings**
  MIT Lincoln Labs. September 2019.
- **AI for Imperfect-Information Game Settings**
  University of Michigan, Ann Arbor. April 2019.
- **AI for Imperfect-Information Game Settings**
  Cornell Tech. April 2019.
- **AI for Imperfect-Information Game Settings**
  University of Texas at Austin. April 2019.
- **AI for Imperfect-Information Game Settings**
  New York University. March 2019.
- **AI for Imperfect-Information Game Settings**
  Cornell University. March 2019.
- **AI for Imperfect-Information Game Settings**
  MIT. March 2019.
- **AI for Imperfect-Information Game Settings**
  Stanford University. February 2019.

- **AI for Imperfect-Information Game Settings**
  University of Maryland, College Park. February 2019.
- **AI for Imperfect-Information Game Settings**
  UMass, Amherst. February 2019.
- **AI for Imperfect-Information Game Settings**
  Duke University. February 2019.

2018

- **Deep Counterfactual Regret Minimization**
  Deep Reinforcement Learning Workshop at NeurIPS Oral Presentation. December 2018.
- **AI for Large Imperfect-Information Games: Beating Top Humans in No-Limit Poker**
  Microsoft Research. October 2018.
- **AI for Large Imperfect-Information Games: Beating Top Humans in No-Limit Poker**
  Google Brain. July 2018.
- **AI for Large Imperfect-Information Games: Beating Top Humans in No-Limit Poker**
  OpenAI. July 2018.
- **AI for Large Imperfect-Information Games: Beating Top Humans in No-Limit Poker**
  Facebook AI Research. June 2018.
- **From Poker AI to Negotiation AI: Dealing with Hidden Information**
  TTI/Vanguard: Intelligence, Natural and Artificial. June 2018.
- **Libratus: Beating Top Pros in No-Limit Poker**
  6th Toulouse Economics and Biology Workshop. May 2018.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Jane Street Capital. January 2018.

2017

- **Safe and Nested Subgame Solving for Imperfect-Information Games**
  NIPS Oral Presentation. December 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Rutgers University. October 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Princeton University. October 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Cubist Systems. October 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Harvard University. September 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  MIT. September 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Hertz Fellowship Retreat. September 2017.
- **Reduced Space and Faster Convergence in Imperfect-Information Games via Pruning**
  ICML Oral Presentation. August 2017.
- **Super-Human AI for Strategic Reasoning: Beating Top Pros in Heads-Up No-Limit Texas Hold'em**
  TNG Big Tech Day. Keynote. June 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Facebook AI Research. May 2017.

- **Libratus: Beating Top Pros in No-Limit Poker**
  DeepMind. May 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  University of Michigan. April 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  Carnegie Mellon University. April 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  OpenAI. February 2017.
- **Libratus: Beating Top Pros in No-Limit Poker**
  AAAI Workshop on Computer Poker and Imperfect-Information Games. February 2017.

2016

- **Reduced Space and Faster Convergence in Imperfect-Information Games via Pruning**
  Carnegie Mellon University. September 2016.
- **Strategy-Based Warm Starting for Regret Minimization in Games**
  AAAI Workshop on Computer Poker and Imperfect-Information Games, 2016.
- **Simultaneous Abstraction and Equilibrium Finding in Games**
  AAAI Workshop on Computer Poker and Imperfect-Information Games, 2016.

2015

- **Simultaneous Abstraction and Equilibrium Finding in Games**
  INFORMS Annual Conference, 2015. Optimization cluster.
- **Simultaneous Abstraction and Equilibrium Finding in Games**
  IJCAI Oral Presentation, 2015.

2014

- **Regret Transfer and Parameter Optimization**
  AAAI Workshop on Computer Poker and Imperfect Information, 2014.
- **Regret Transfer and Parameter Optimization**
  AAAI Oral Presentation, 2014.

Teaching

Fall 2017	**Teaching Assistant**, *Artificial Intelligence*, Carnegie Mellon University
Fall 2014	**Teaching Assistant**, *Artificial Intelligence*, Carnegie Mellon University
Spring 2009	**Recitation Instructor**, *Calculus I*, Rutgers University
Fall 2008	**Recitation Instructor**, *Calculus I*, Rutgers University
Spring 2008	**Recitation Instructor**, *Pre-Calculus*, Rutgers University
Fall 2007	**Recitation Instructor**, *Pre-Calculus*, Rutgers University
Spring 2007	**Peer Mentor**, *Calculus II*, Rutgers University
Fall 2006	**Peer Mentor**, *Calculus II*, Rutgers University
Summer 2006	**Teaching Assistant**, *Rutgers Young Scholars Program*, Rutgers University

Outreach

Summers 2018–2023 **Instructor**, *Rutgers Young Scholars Program*, Rutgers University
Taught a week-long intensive course on game theory to gifted high school students

Summers 2009–2017 **Guest Lecturer**, *Rutgers Young Scholars Program*, Rutgers University
Presented lessons on game theory to gifted high school students

2015–2018 **Presenter**, *Creative Technologies Nights*, Carnegie Mellon University
Introduce middle school girls to various technology-related topics in weekly presentations as part of a program to encourage young women to pursue higher education and careers in STEM fields.

2010–2012 **Organizer and Presenter**, *FedEd*, Federal Reserve Board of Governors
Taught financial literacy and monetary policy through guest lectures to D.C.-area public and private high school students.

2008–2009 **Presenter**, *Douglas Project: Women in STEM*, Rutgers University
Presented lessons on game theory to gifted high school girls as part of a program to encourage young women to pursue higher education and careers in STEM fields.